

Lab15 – Service types

In this exercise, you will create a Deployment and expose a container port for its Pods. You will demonstrate the differences between the service types ClusterIP and NodePort.

1. Create a Service named `myapp` of type `ClusterIP` that exposes port 80 and maps to the target port 80.

```
vagrant@kubernetes-control-plane:~/lab10$ kubectl create service clusterip myapp --tcp=80:80
service/myapp created
vagrant@kubernetes-control-plane:~/lab10$ kubectl get services
NAME         TYPE        CLUSTER-IP    EXTERNAL-IP    PORT(S)    AGE
kubernetes   ClusterIP   10.96.0.1     <none>         443/TCP    23m
myapp        ClusterIP   10.97.234.12  <none>         80/TCP     12s
vagrant@kubernetes-control-plane:~/lab10$
```

2. Create a Deployment named `myapp` that creates 1 replica running the image `nginx:1.23.4-alpine`. Expose the container port 80.

```
vagrant@kubernetes-control-plane:~/lab10$ kubectl create deployment myapp --image=nginx:1.23.4-alpine --port=80
deployment.apps/myapp created
vagrant@kubernetes-control-plane:~/lab10$ kubectl get deployments,pods
NAME                READY   UP-TO-DATE   AVAILABLE   AGE
deployment.apps/myapp  0/1     1             0           9s

NAME                READY   STATUS              RESTARTS   AGE
pod/myapp-64fb67cd9-r8f8l  0/1     ContainerCreating   0           9s
pod/pod1              1/1     Running             0           20m
vagrant@kubernetes-control-plane:~/lab10$
```

3. Scale the Deployment to 2 replicas.

```
vagrant@kubernetes-control-plane:~/lab10$ kubectl scale deployment myapp --replicas=2
deployment.apps/myapp scaled
vagrant@kubernetes-control-plane:~/lab10$ kubectl get deployments,pods
NAME                READY   UP-TO-DATE   AVAILABLE   AGE
deployment.apps/myapp  2/2     2             2           24m

NAME                READY   STATUS              RESTARTS   AGE
pod/myapp-64fb67cd9-hsbk5  1/1     Running             0           12s
pod/myapp-64fb67cd9-r8f8l  1/1     Running             0           24m
pod/pod1              1/1     Running             0           44m
vagrant@kubernetes-control-plane:~/lab10$
```

4. Create a temporary Pod using the image `busybox` and run a `wget` command against the IP of the service.

Determine the cluster IP and the port for the Service. In this case, it's `10.97.234.12:80`. Alternatively, you can use the DNS name `myapp`. Use the information with the `wget` command.

```
vagrant@kubernetes-control-plane:~/lab10$ kubectl run tmp --image=busybox --restart=Never -it --rm -- wget -O- 10.97.234.12:80
Connecting to 10.97.234.12:80 (10.97.234.12:80)
writing to stdout
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, the nginx web server is successfully installed and
working. Further configuration is required.</p>

<p>For online documentation and support please refer to
<a href="http://nginx.org/">nginx.org</a>.<br/>
Commercial support is available at
<a href="http://nginx.com/">nginx.com</a>.</p>

<p><em>Thank you for using nginx.</em></p>
</body>
</html>
-          100% |*****| 615 0:00:00 ETA
written to stdout
pod "tmp" deleted
vagrant@kubernetes-control-plane:~/lab10$
```

5. Change the service type so that the Pods can be reached from outside of the cluster. You must turn the type of the service into `NodePort` to expose it outside of the cluster. Now, the service should expose a port in the 30000 range.

```
ports:
- name: 80-80
  port: 80
  protocol: TCP
  targetPort: 80
selector:
  app: myapp
sessionAffinity: None
type: NodePort
status:
```

```
vagrant@kubernetes-control-plane:~/lab10$ kubectl edit service myapp
service/myapp edited
vagrant@kubernetes-control-plane:~/lab10$
vagrant@kubernetes-control-plane:~/lab10$ kubectl get services
NAME         TYPE          CLUSTER-IP    EXTERNAL-IP    PORT(S)          AGE
kubernetes   ClusterIP     10.96.0.1     <none>         443/TCP          58m
myapp        NodePort      10.97.234.12  <none>         80:31605/TCP     35m
vagrant@kubernetes-control-plane:~/lab10$
vagrant@kubernetes-control-plane:~/lab10$
```

6. Run a browser or `wget` command against the service from outside of the cluster.

